Lecture 2: Basic of light and solar spectrum

- The relationship between the bandgap of a material and the absorption range of the material
- The concept of multi-junction solar cell (what are the advantages?)
- Calculation of power density from photon flux (Φ) and spectral irradiance (S)
- Power density calculation from black body radiation
- Power density calculation on an object some distance away from the sun
- Sun position calculation and Lab 1 Quiz (need to understand zenith angle, azimuth angle, tilted angle, incident angle, air mass etc.)

Lecture 3: Semiconductor and P-N junction

- Concept of absorption coefficient, absorption spectrum, absorption depth, Lambert-Beer Law
- Bulk and surface recombination in semiconductors
- Reverse situation current (J_0), dark current, diode current equation, diode ideality factor

Lecture 4 – Solar Cell Operation

- Electron-hole separation and photocurrent generation in p-n junction solar cell
- Charge generation and charge collection probability vs distance in the device
- The effect of surface and rear recombination, diffusion length on charge collection probability
- The effect of surface and rear recombination, diffusion length, reflection etc. on EQE spectrum
- Jsc calculation from EQE spectrum

Lecture 4 – Solar Cell Operation

- J-V characteristic of a solar cell, electrical parameters extraction (Jsc, Voc, Rsh, Rs), and power conversion efficiency calculation
- Effect of Rs and Rsh on the J-V curve
- Equivalent circuit diagram with and without parasitic resistance
- Voc calculation based on reverse situation current (I_0)
- Effect of temperature and light intensity on the performance of a solar cell

Lecture 5 – Crystalline Silicon Solar cells

- Difference between the monocrystalline and multicrystalline silicon solar cells
- Techniques to improve the absorption of silicon solar cell
- Calculation of the thickness and refractive index of anti-reflection coating layer
- Doping optimization (trade-off between the diffusion length and Voc)
- Must understand the purpose (function) of each layer (features) in silicon solar cell
- Various structures of silicon solar cells (screen-printed solar cell, PERC, PERL, buried contact, interdigitated back contact, etc.)
- Fabrication process of screen-printed solar cell

Lecture 6 – Organic solar cell

- Understanding of electron delocalization in organic semiconductor (sp3 hybridization and p_z orbital overlapping)
- Differences between normal polymer and conjugated polymer, and underlying mechanism
- Components of typical conjugated polymers and their roles in organic solar cells
- Properties of organic semiconductor (absorption coefficient, exciton binding energy, exciton diffusion length, mobility, isotropic charge transport etc.) and their influence on organic solar cells

- Anisotropic charge transport in conjugated polymer
- Working principle of organic solar cell
- Energy levels (LUMO, HOMO) of donor and acceptor and their influence on Voc
- Work function of electrodes and their influence on Voc
- Various device architectures of organic solar cells, their advantages, and disadvantages
- Conventional structure vs Inverted structure and their charge collection nature

Lecture 7 – Perovskite solar cell

- Perovskite structure
- Formation of 3D and 2D perovskite (Must understand unit cell structures and their general formula)
- Advantages and disadvantages of 2D perovskite
- Tolerance Factor calculation and structure stability
- Composition of ions and bandgap tuning
- Optoelectronic properties of perovskite (absorption, exciton binding energy, charge mobility, diffusion length, etc.)
- Bulk and surface defects in perovskite, their formation energy
- Various structures of perovskite solar cells

Other Tips

- No complicated calculation, but a simple calculation is required.
- Some questions need explanation. Don’t worry about language (English) for those questions. Keywords are the most important. You will not be penalized for language errors.
- You may also use the short form to describe your explanation.
- For example, for the statement “ The current density increases with temperature due to the reduced bandgap at higher temperatures” you can simply write as below.
- $T\uparrow \;\rightarrow\; J_{sc}\uparrow\ (\because E_g\downarrow)$

- If you see words such as “illustrate”, “sketch,” etc., you must draw a figure to explain.
- Don’t spend too much time on one question. There are 4 questions, and you should not spend more than 30 minutes on each.
- Read the questions carefully. The questions are long, but you can get the answers quickly if you understand the concepts.